MODULE 14, LESSON 3

AI 101 FOR MECHANICAL ENGINEERS

Tools act. Agents coordinate.

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Lesson Structure

1. System Components
2. Integration and Verification
3. Course Operating Environment

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Section 1

System Components

Defining the functional blocks of automated software systems.

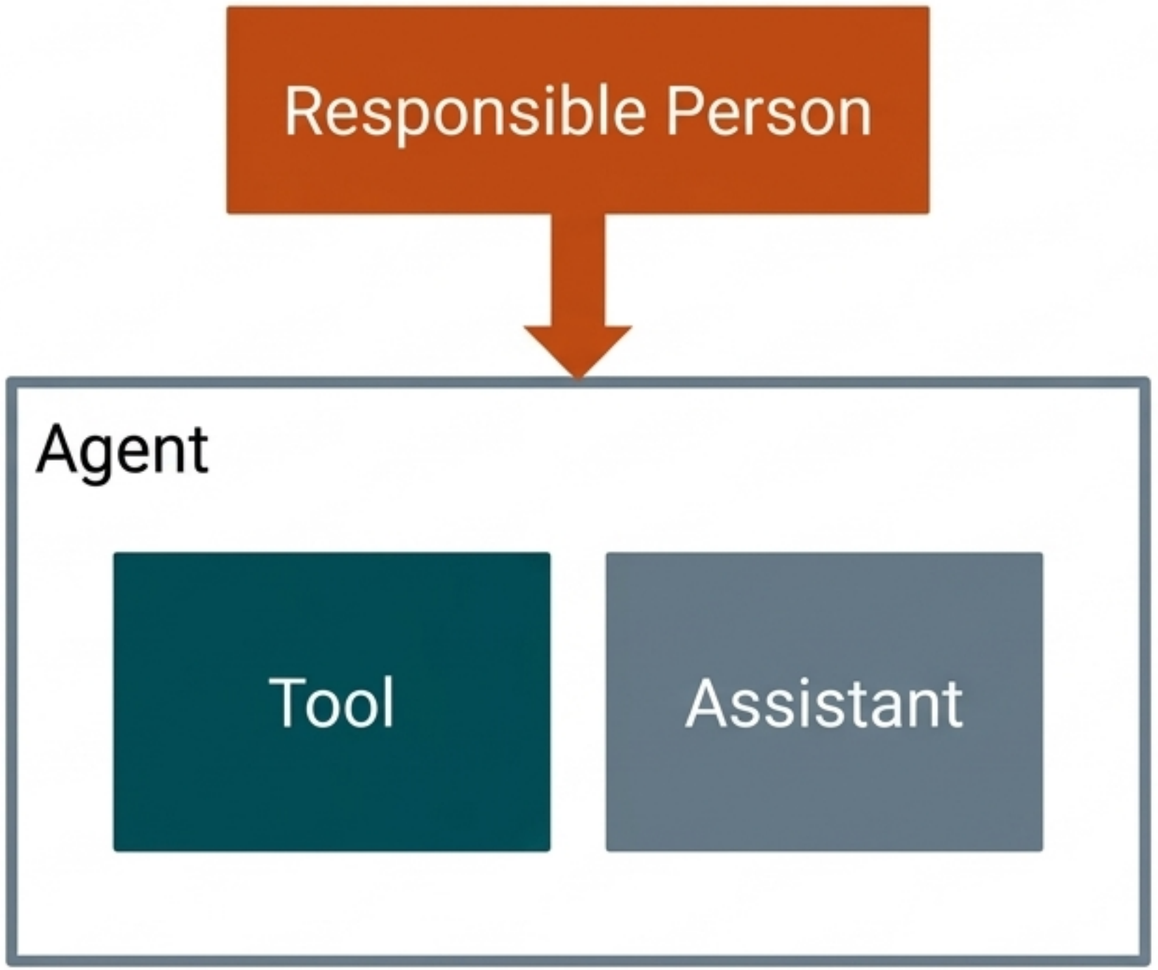
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System Component Matrix

Assistant	Tool	API	Agent
■ Interface supporting user tasks	■ Performs a specific operation	■ Defined software interface	■ Coordinates actions toward goals
Conversational engineering helper	Run Python or search file	Submit data via documented function	Bounded workflow with permissions

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System Coordination Hierarchy



Agents select actions with defined autonomy limits.

Assistants propose actions to system users.

Tools carry out the specific proposed operations.

A responsible person dictates permissions and logs.

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Section 2

Integration and Verification

Separating software code generation from verified engineering execution.

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Execution Divergence



Writing code differs fundamentally from executing code.

Code execution without errors remains functionally incomplete.

Successful execution never guarantees correct engineering results.

Generated outputs demand rigorous physical system verification.

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The Verification Pipeline



Assistants draft code for proposed operational actions.

Tools apply the approved code to systems.

Strict intermediate checks must isolate both actions.

Real systems require independent responsible person approval.

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Mandatory Pre-Execution Checks



Verify software syntax and independent logic tests.



Audit system action permissions and execution logs.



Confirm mathematical equations and precise SI units.



Obtain explicit approval from the responsible person.



Validate operational domains and exact input limits.

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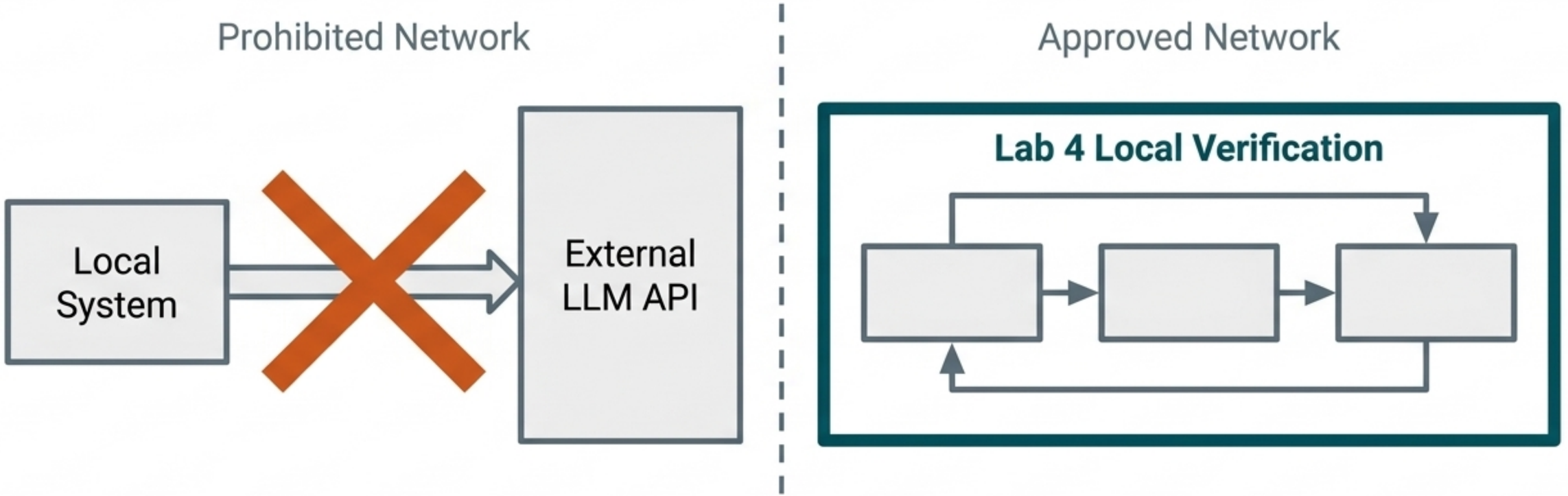
Section 3

Course Operating Environment

Defining the offline boundaries of the laboratory prompt builder.

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Offline Prompt Builder Architecture



The course prompt builder operates entirely offline.

The system contains no connected AI services.

The builder generates text for independent review.

Lab 4 verifies illustrative drafts strictly locally.

The environment never initiates network LLM calls.

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Key Points

1. Agents coordinate tools and assistants within bounds.
2. Successful execution demands strict physical constraint verification.
3. Course laboratory tools operate locally without networks.

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